

Introduction to the Rust Programming Language

2026-10-01 — A Purdue Hackers workshop

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Ferris, the Rust mascot

What is Rust?



What is Rust?

Rust is a general-purpose programming language that empowers everyone to build reliable and efficient software.

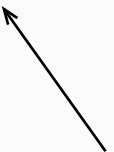
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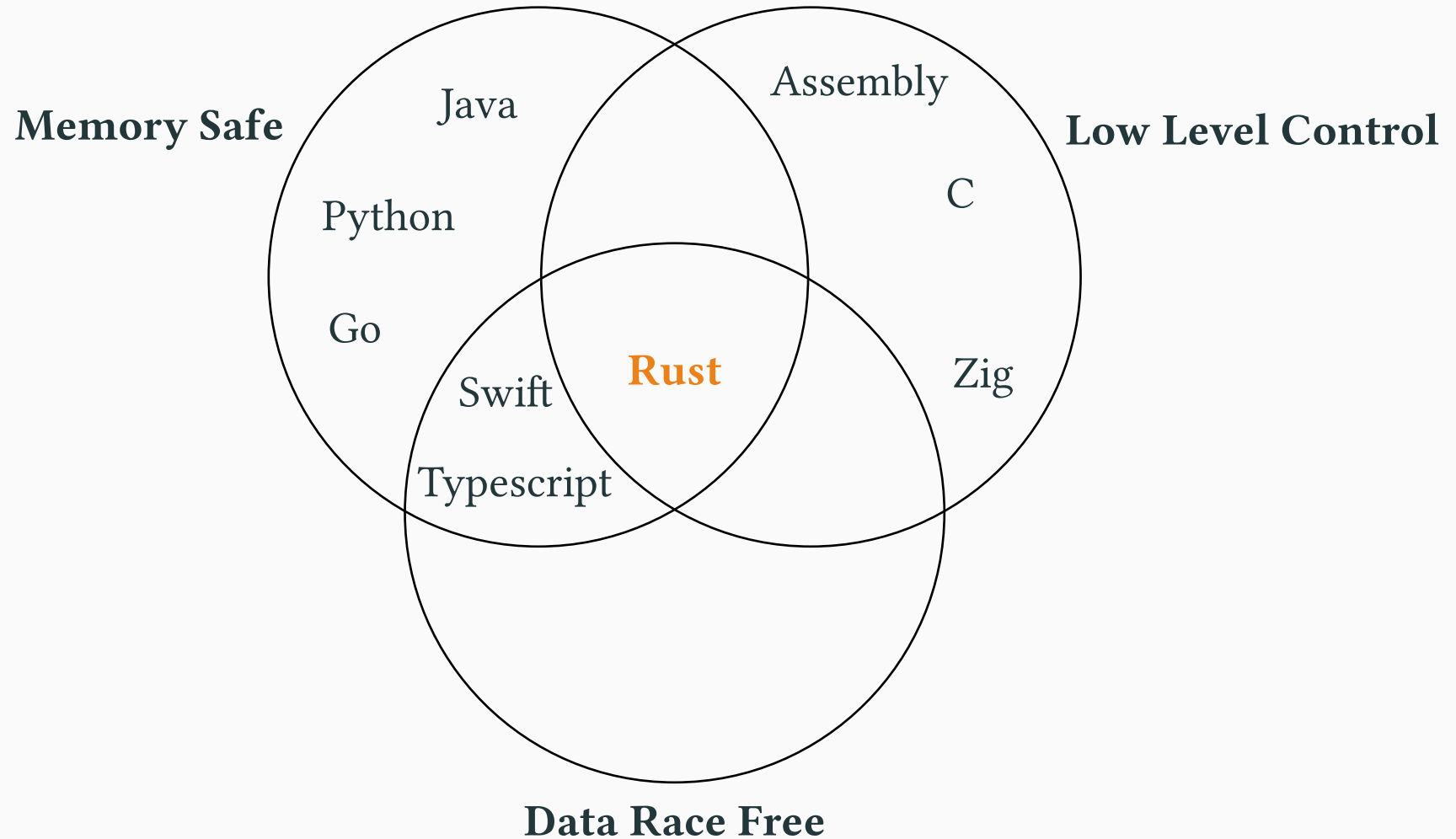
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How?

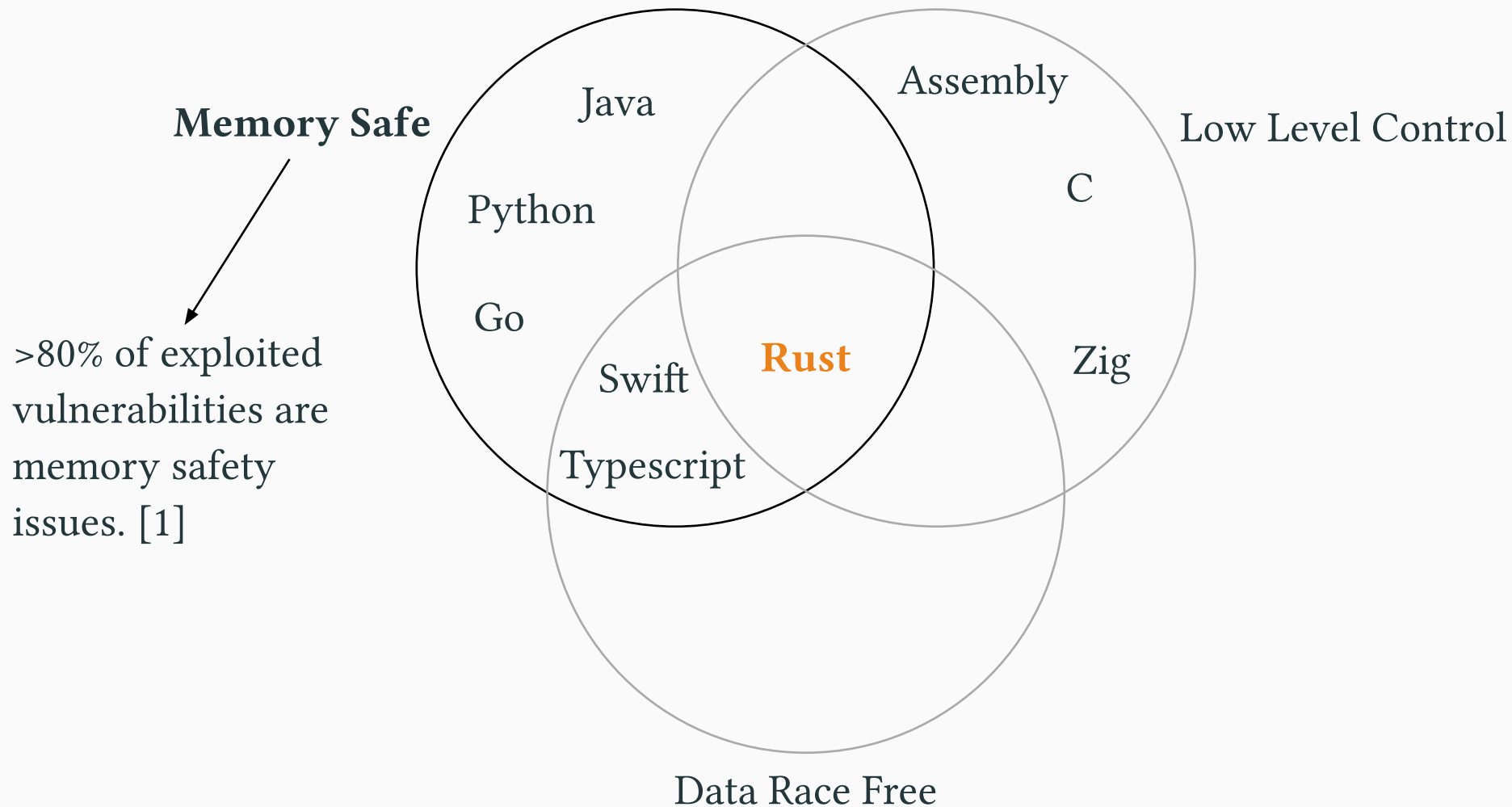


How is Rust empowering?

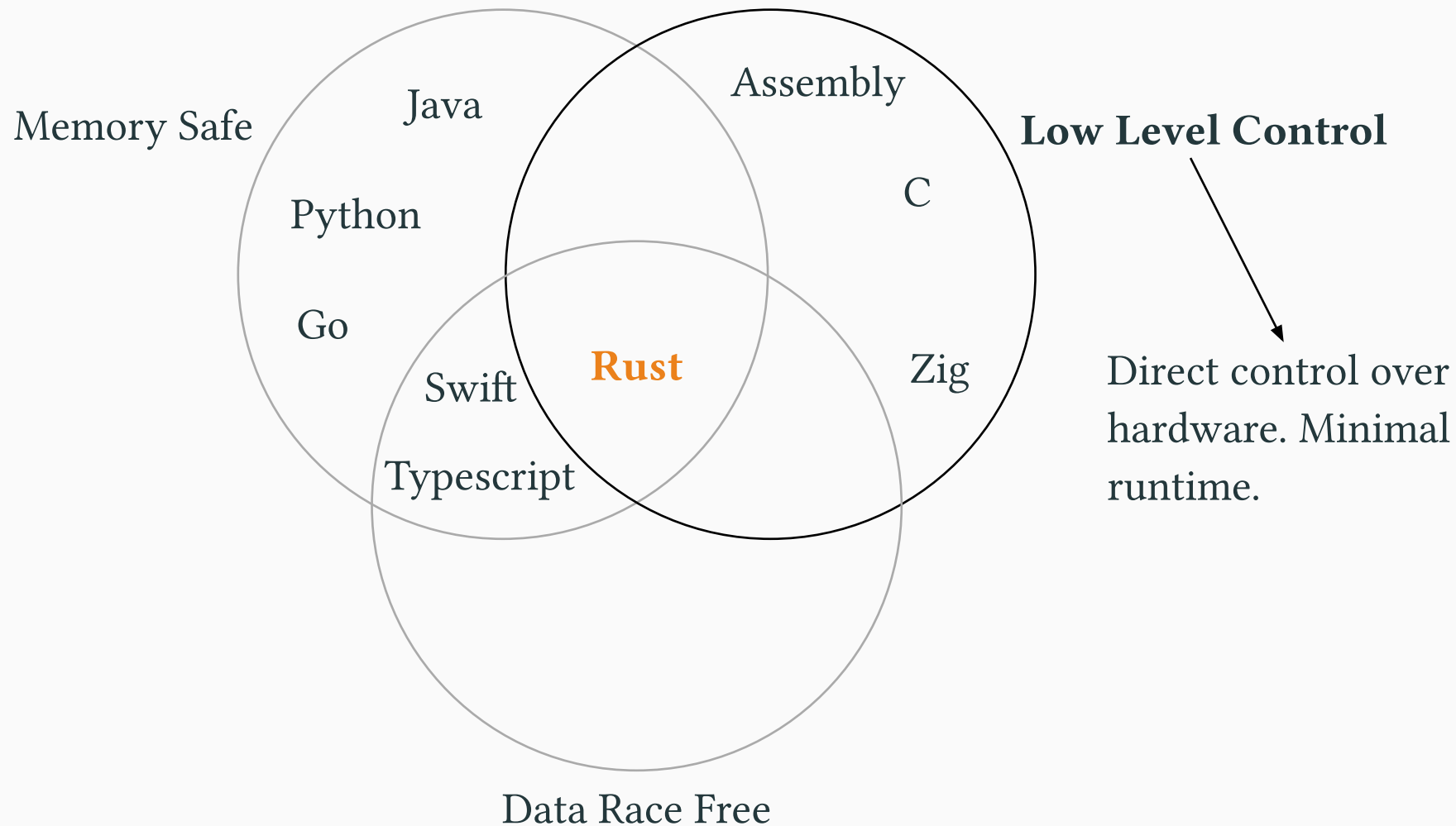
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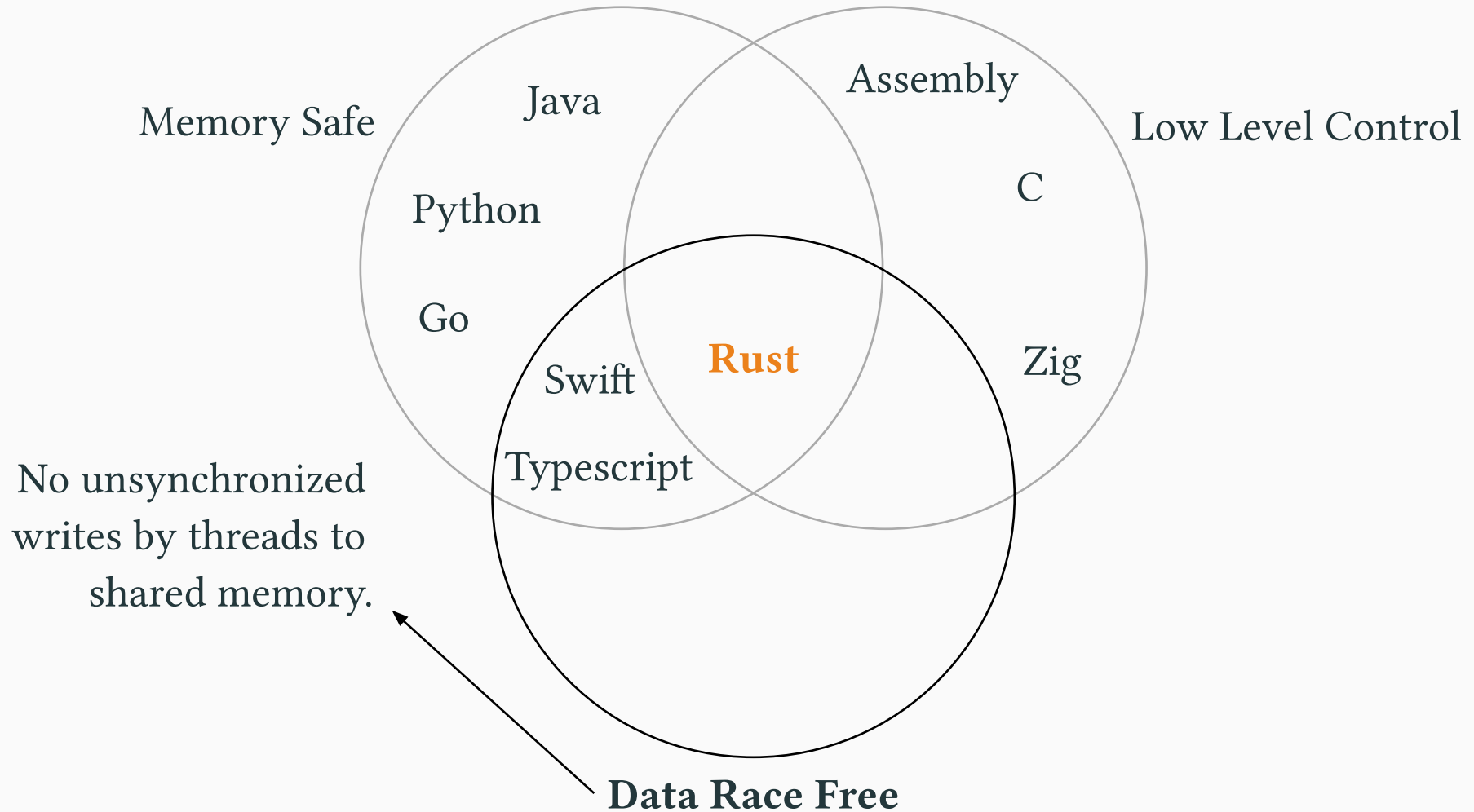
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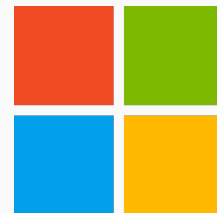
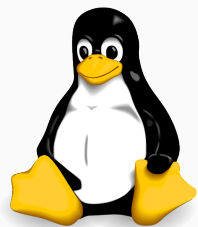
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Who uses Rust?

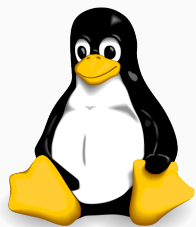
Who uses Rust?

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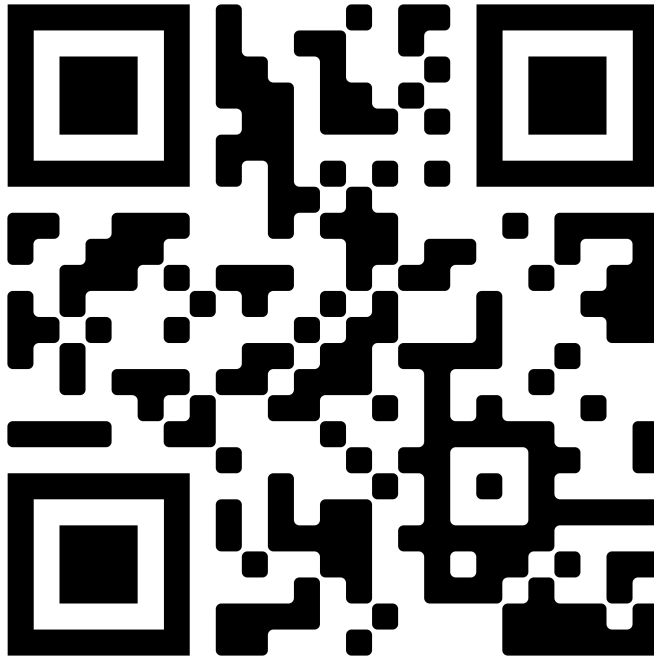


As do AI agents



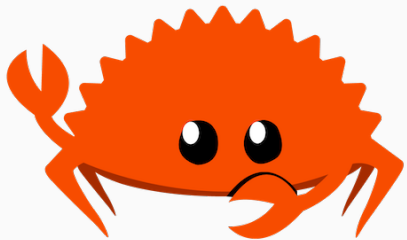
Let's learn some Rust!

<https://code.purduehackers.com/new/purduehackers/rust-workshop-2026>



Hello World

```
fn main() {  
    println!("Hello World!");  
}
```



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Ownership

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Ownership

Let's focus on a very common data structure: strings

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fn main() {  
    let ph: &str = "Purdue Hackers";  
    println!("{ph} is awesome!");  
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“Purdue Hackers” is hardcoded into the program

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fn main() {  
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Rust program ->

Thank you for listening!

Appendix

Bibliography

- [1] “What is memory safety and why does it matter?,” *Internet Security Research Group*. [Online]. Available: <https://www.memorysafety.org/docs/memory-safety/>